Module 10-5 to 10-7 #1

Student Name: _____

Class Name: **ALEKS Integrated Mathematics II**

Number of Questions: **29**

Instructor Name: **Bodanske, Daniel**

Question 1 of 29

Simplify.

$$\sqrt{w^{36}}$$

Assume that the variable represents a positive real number.

Question 2 of 29

Simplify.

$$\sqrt{16y^4}$$

Assume that the variable y represents a positive real number.

Question 3 of 29

Find the value of $\sqrt[3]{-125}$.

Question 4 of 29

Simplify.

$$\sqrt[3]{125u^{15}}$$

Assume that the variable represents a positive real number.

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Question 5 of 29

Simplify.

$$\sqrt{63}$$

Question 6 of 29

Simplify.

$$\sqrt{189}$$

Question 7 of 29

Simplify.

$$\sqrt{45u^8}$$

Assume that the variable u represents a positive real number.

Question 8 of 29

Simplify.

$$\sqrt{81v^{11}}$$

Assume that the variable represents a positive real number.

Question 9 of 29

Simplify.

$$\sqrt{64x^3y^7}$$

Assume that all variables represent positive real numbers.

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Question 10 of 29

Write the following in simplified radical form.

$$\sqrt[4]{80}$$

Question 11 of 29

Write the following in simplified radical form.

$$\sqrt[4]{81u^{11}}$$

Assume that the variable represents a positive real number.

Question 12 of 29

Write the following expression in simplified radical form.

$$\sqrt[4]{256t^{11}w^{5}}$$

Assume that all of the variables in the expression represent positive real numbers.

Question 13 of 29

Simplify.

$$\sqrt{5a^{5}}\sqrt{20a^{2}}$$

Assume that the variable represents a positive real number.

Question 14 of 29

Simplify.

$$\sqrt[4]{3}\cdot\sqrt[4]{8}$$

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Question 15 of 29

Simplify.

$$\sqrt[4]{2y^6} \cdot \sqrt[4]{8y^5}$$

Assume that the variable represents a positive real number.

Question 16 of 29

Rationalize the denominator and simplify.

$$\sqrt{\frac{3}{2}}$$

Question 17 of 29

Simplify.

$$-\sqrt{8} - 3\sqrt{32}$$

Question 18 of 29

Simplify.

$$5\sqrt{63} + \sqrt{28} + 17\sqrt{7}$$

Question 19 of 29

Multiply.

$$\sqrt{5}\left(4 + \sqrt{7}\right)$$

Simplify your answer as much as possible.

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Question 20 of 29

Multiply.

$$8\sqrt{6}\left(\sqrt{6}-\sqrt{7}\right)$$

Simplify your answer as much as possible.

Question 21 of 29

Multiply.

$$\left(3\sqrt{2}+1\right)\left(3+2\sqrt{10}\right)$$

Simplify your answer as much as possible.

Question 22 of 29

If an object is dropped from a height of h meters and hits the ground in t seconds, then $t=\sqrt{\frac{h}{4.9}}$.

Suppose that an object is dropped from the top of a building that is 463.54 meters tall. How long does it take to hit the ground?

Round your answer to the nearest tenth.

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Question 23 of 29

Classify each number below as an integer or not.

	Integer?	
	Yes	**No**
14.37	○	○
$\frac{24}{6}$	○	○
$\frac{7}{2}$	○	○
-11	○	○
955.87	○	○

Question 24 of 29

For each sum or product, determine whether the result is a rational number or an irrational number.

Then choose the appropriate reason for each.

A: The sum of two rationals is always rational.
B: The sum of a rational and an irrational is always irrational.
C: The sum of two irrationals is sometimes rational, sometimes irrational.
D: The product of two rationals is always rational.
E: The product of a nonzero rational and an irrational is always irrational.
F: The product of two irrationals is sometimes rational, sometimes irrational.

	Result is	**Reason**
(a) $\frac{11}{13} + \frac{4}{15}$	{ rational , irrational }	{ A , B , C , D , E , F }
(b) $29 + \sqrt{10}$	{ rational , irrational }	{ A , B , C , D , E , F }
(c) $37 \times \frac{7}{8}$	{ rational , irrational }	{ A , B , C , D , E , F }
(d) $\sqrt{22} \times 3$	{ rational , irrational }	{ A , B , C , D , E , F }

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Question 25 of 29

Solve for v, where v is a real number.

$$\sqrt{v} = -6$$

Question 26 of 29

Solve for x, where x is a real number.

$$-5 + \sqrt{x - 9} = 4$$

If there is more than one solution, separate them with commas.
If there is no solution, click on "No solution".

$x =$

Question 27 of 29

If a person's eye level is h meters above sea level and she can see d kilometers to the horizon, then $d = 3.6\sqrt{h}$. Suppose the person can see 15.3 kilometers to the horizon. What is the height of her eye level above sea level?

Carry your intermediate computations to at least four decimal places, and round your answer to the nearest tenth.

Question 28 of 29

Solve for x.

$$81^{5-3x} = 9^{-4x}$$

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Question 29 of 29

Multiply.

$$(x-10)(x+10)$$

Simplify your answer.

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Class Name: **ALEKS Integrated Mathematics II** Number of Questions: **29**

Question 1 of 29

w^{18}

Question 2 of 29

$4y^2$

Question 3 of 29

-5

Question 4 of 29

$5u^5$

Question 5 of 29

$3\sqrt{7}$

Question 6 of 29

$3\sqrt{21}$

Question 7 of 29

$3u^4\sqrt{5}$

Question 8 of 29

$9v^5\sqrt{v}$

Question 9 of 29

$8xy^3\sqrt{xy}$

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Question 10 of 29

$2\sqrt[4]{5}$

Question 11 of 29

$3u^2\sqrt[4]{u^3}$

Question 12 of 29

$4t^2 w\sqrt[4]{t^3 w}$

Question 13 of 29

$10a^3\sqrt{a}$

Question 14 of 29

$\sqrt[4]{24}$

Question 15 of 29

$2y^2\sqrt[4]{y^3}$

Question 16 of 29

$\frac{\sqrt{6}}{2}$

Question 17 of 29

$\mathbf{-14\sqrt{2}}$.

Question 18 of 29

$34\sqrt{7}$

Question 19 of 29

$4\sqrt{5}+\sqrt{35}$

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Question 20 of 29

$48 - 8\sqrt{42}$

Question 21 of 29

$9\sqrt{2} + 12\sqrt{5} + 3 + 2\sqrt{10}$

Question 22 of 29

9.7 seconds

Question 23 of 29

	Integer?	
	Yes	**No**
14.37	○	◉
$\frac{24}{6}$	◉	○
$\frac{7}{2}$	○	◉
-11	◉	○
955.87	○	◉

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Question 24 of 29

A: The sum of two rationals is always rational.
B: The sum of a rational and an irrational is always irrational.
C: The sum of two irrationals is sometimes rational, sometimes irrational.
D: The product of two rationals is always rational.
E: The product of a nonzero rational and an irrational is always irrational.
F: The product of two irrationals is sometimes rational, sometimes irrational.

	Result is	**Reason**
(a) $\frac{11}{13} + \frac{4}{15}$	rational	A
(b) $29 + \sqrt{10}$	irrational	B
(c) $37 \times \frac{7}{8}$	rational	D
(d) $\sqrt{22} \times 3$	irrational	E

Question 25 of 29

No solution

Question 26 of 29

$x = 90$

Question 27 of 29

18.1 meters

Question 28 of 29

$x = 5$

Question 29 of 29

$x^2 - 100$

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